The atmosphere as a stack of regimes

What is the thing at 90 to 300 km, and which of its properties set instrument requirements?

`Notebook companion: Unit-02-atmosphere.ipynb`

1 The question

Unit 1 established why the density of the upper atmosphere is worth measuring. This unit is about what that atmosphere actually is.

The answer is not one thing. Over the range the instrument reaches, the atmosphere passes through three regimes. At the bottom it is a stirred mixture with the same composition at every height. Above that it is a set of separate gases, each settling under its own weight. Higher still it is so thin that the description of a gas in equilibrium begins to fail.

Three properties of those regimes set instrument requirements directly. The rate at which density falls with height determines how much more atmosphere a limb path traverses than a vertical one. The relative abundance of the species determines which of them can be measured at which altitudes. The shape of the temperature profile determines how far a two-parameter model of it can be trusted.

2 Physical picture

Two ideas are needed for the whole unit.

The first is hydrostatic balance. The pressure at any height supports the weight of everything above it. Combining that condition with the ideal gas law gives a density that falls exponentially with a characteristic distance

$$H = \frac{kT}{mg},$$

the scale height. It is the distance over which the density falls by a factor of e. At these altitudes it is tens of kilometres rather than the eight kilometres familiar at the ground, because the thermosphere is hot and the gas that remains is light.

The second idea is what the mass m in that expression refers to, and it is what produces the regimes. Turbulent mixing moves parcels of gas bodily, so it transports all species at the same rate whatever their individual masses. Where mixing dominates, every species therefore falls off on the scale height computed from the *mean* mass, and the composition is the same at every height. Molecular diffusion moves individual molecules, and a heavier molecule diffuses more slowly, so the rate does depend on mass. Where diffusion dominates, each species falls off on the scale height computed from its *own* mass, and the lighter species come to dominate with height. The boundary between the two regimes, the turbopause, sits near 100 km.

Above the turbopause the composition therefore changes continuously with altitude. Atomic oxygen, at 16 atomic mass units, falls off nearly twice as slowly as molecular nitrogen at 28, so oxygen

overtakes nitrogen and eventually dominates. That changing ratio is what makes composition measurable at all: two species in a fixed ratio could not be separated by absorption measurements that respond to their sum. Section 4d shows that the same changing ratio is why nitrogen ceases to be measurable at the top of the instrument's range.

Higher still, the gas becomes so thin that a molecule travels further between collisions than the distance over which the density changes. Above that height, the exobase, the atmosphere is no longer a fluid in equilibrium and individual particles follow ballistic trajectories.

3 Derivation

Hydrostatic balance says $dP/dz = -\rho g$. With the ideal gas law $P = nkT$ and $\rho = nm$,

$$\frac{1}{n}\frac{dn}{dz} = -\frac{mg}{kT} - \frac{1}{T}\frac{dT}{dz},$$

so the local scale height read off a measured profile is

$$H_{\text{meas}} \equiv -\left(\frac{d\ln n}{dz}\right)^{-1},$$

which equals kT/mg only where the temperature is constant. In the lower thermosphere the temperature rises steeply with height, so the second term above is not negligible. That is one reason the measured and predicted scale heights in section 4b disagree there.

The mass to use depends on the regime. Where mixing dominates, all species share the mean mass $\bar{m}$ and the profile of every species falls on $kT/\bar{m}g$. Where diffusion dominates, each species s falls on its own $kT/m_s g$. Comparing a measured scale height with both predictions is therefore a direct test of which regime the atmosphere is in — and section 4b shows why that test is harder to run than it looks.

4 Worked numbers

4a. Composition against height

Altitude	Temperature	Mean mass	O/N_2
95 km	185 K	28.66 amu	0.03
110 km	234 K	27.14 amu	0.17
140 km	603 K	23.77 amu	0.66
200 km	805 K	20.10 amu	2.04
300 km	837 K	16.93 amu	9.91
500 km	838 K	13.14 amu	196

Two things to read from the table. First, the mean molecular mass is already falling at 95 km, the bottom of this grid. Diffusive separation is what makes it fall, so the turbopause sits at or below the lowest altitude the instrument reaches, and every altitude the instrument measures is either

diffusively separated or in the transition to it. Second, atomic oxygen overtakes molecular nitrogen at 159 km, and above that the ratio grows quickly: a factor of ten by 300 km and two hundred by 500.

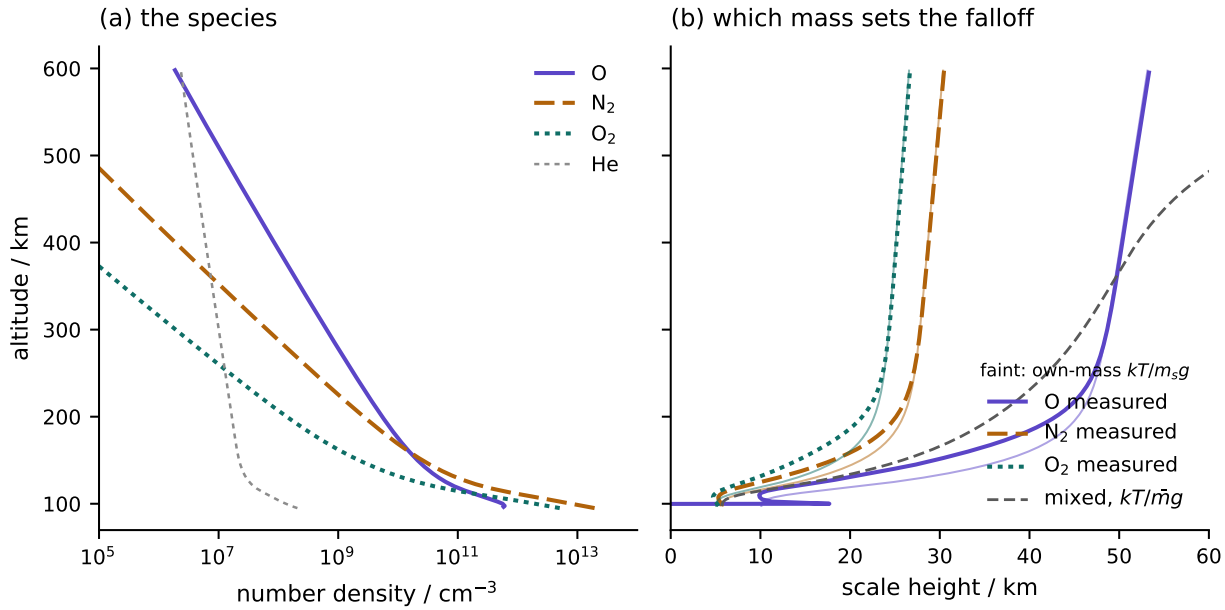


Figure 1: (a) Number density of the major species, with helium shown because it dominates the mass above about 500 km and is omitted from the instrument's optical model. (b) The scale height measured from each profile, against the two predictions: each species' own kT/m_sg in faint lines and the mixed-atmosphere $kT/\bar{m}g$ dashed. Above about 250 km each measured curve lies on its own prediction. Generated by section B of the notebook.

4b. Testing the regime, and what the test cannot do

z	$\bar{m}$	O		N_2		O_2		mixed
		meas	own	meas	own	meas	own	
100	28.21	17.7	10.1	5.6	5.8	5.2	5.1	5.7
120	25.79	13.5	21.0	8.1	12.0	6.8	10.5	13.0
160	22.32	33.2	39.9	20.2	22.8	16.1	19.9	28.6
200	20.10	42.9	45.4	25.1	25.9	21.4	22.7	36.1
300	16.93	48.7	48.6	27.8	27.8	24.4	24.3	46.0
400	15.43	50.3	50.2	28.8	28.7	25.2	25.1	52.0

Scale heights in km. "meas" is read off the profile; "own" is kT/m_sg; "mixed" is $kT/\bar{m}g$. Mean mass in atomic mass units.

Above 300 km the test is unambiguous. Every measured scale height sits on its own species' value to within a fraction of a per cent, and all three are far from the mixed-atmosphere value. The atmosphere there is diffusively separated, and the retrieval is entitled to assume it.

Below 150 km the test returns no verdict, for two reasons that are worth stating plainly.

The first is that the two species available for the test have masses too close to the mean mass. Molecular nitrogen has a mass of 28 atomic mass units and the mean mass near 100 km is 28.2, so its own scale height and the mixed-atmosphere scale height differ by about two per cent. That is smaller than the uncertainty in reading a scale height off a profile, so a nitrogen measurement cannot separate the two regimes. Molecular oxygen, at 32 atomic mass units, is only slightly further from the mean mass and is barely better.

The second is that the one species whose mass is far enough from the mean for the test to be decisive, atomic oxygen, matches neither prediction. Its measured scale height at 100 km is 17.7 km, against 10.1 for diffusive separation and 5.7 for mixing. Atomic oxygen near 100 km is in neither equilibrium: it is produced by photodissociation above and destroyed by three-body recombination below, so its profile is set by chemistry rather than by transport.

The conclusion is that the turbopause is better located from the mean mass, which falls steadily from 95 km upward, than from a scale-height test. The scale-height test is useful higher up, where it confirms the diffusive separation that the retrieval assumes.

4c. How far one analytic temperature profile reaches

The retrieval represents the temperature by the Bates form, which rises from a fixed value at 120 km towards an exospheric asymptote with two free parameters. The question is not whether it fits the temperature but whether the density structure it implies is accurate, and over what range.

Fit floor	T_∞	**RMS** T	T at 120 km	**O at 130**	**N_2 at 130**	**O at 400**
100 km	843 K	21.4 K	−21.4 K	+24.4 %	+24.6 %	−4.4 %
110 km	839 K	7.0 K	−21.4 K	+12.8 %	+6.6 %	−1.2 %
130 km	838 K	1.7 K	−21.4 K	+10.9 %	+3.8 %	−0.8 %
170 km	838 K	0.3 K	−21.4 K	+12.4 %	+6.0 %	−1.2 %
190 km	838 K	0.1 K	−21.4 K	+12.8 %	+6.6 %	−1.3 %

Density errors are for a diffusive profile anchored at the 200 km reference, so 200 km is exact by construction and the columns show how the shape propagates away from it.

The temperature fit is close as soon as the mesosphere is excluded: a root-mean-square error of 1.7 K with a floor at 130 km, falling to 0.1 K at 190. The density profile implied by that temperature fit is not close at the bottom of the range. Whatever the fit floor, the profile is about 21 K too cold at 120 km, because the Bates lower boundary is fixed at 360 K while the model atmosphere is at 381 K there, and the resulting density error at 130 km is 11 to 13 per cent in oxygen. Above 200 km the same profile is good to one or two per cent.

WHY NOT a free temperature profile

Because the model is accurate over the range where it is used. Above 200 km *the Bates form reproduces the density structure to one or two per cent, which is well inside the ten per cent accuracy target, and it does so with two free parameters rather than the many that a free profile would require and that the measurement cannot constrain. Below 150* km *it is 11 to 13 per cent wrong in oxygen no matter where it is fitted, and the retrieval therefore does not claim that range. The domain of validity is a measured property of the model, not a convenience.*

4d. Three instrument properties the atmosphere fixes

Where nitrogen stops being measurable. Unit 6 finds the nitrogen slant column determined to better than ten per cent only up to 260 km, and treats that as a limit of the channel set. It is not; it is a property of the atmosphere. Nitrogen is 45 per cent of the two absorbing species at 170 km, 33 per cent at 200, 18 per cent at 250 and 9 per cent at 300. Above the crossover a channel measuring total absorption is measuring mostly oxygen, and no filter choice changes that.

What the limb geometry gains. The limb enhancement derived in Unit 5 is $\sqrt{2\pi(R_E + z)/H}$, so it is set by the scale heights computed in this unit. With H for atomic oxygen of 37.1 km at 150 km rising to 48.6 km at 300, the enhancement falls from 33.2 to 29.4. The limb advantage is largest exactly where the atmosphere is coldest and most compressed.

Where the fluid description ends. Taking a kinetic collision cross section of 3×10^{-15} cm^2, the mean free path is 0.05 km at 150 km and 3.3 km at 300 km, and it reaches the scale height at 443 km. Below that the gas is collisional and hydrostatic balance is meaningful; above it the concept of a local temperature weakens, which is one reason the retrieval does not extend indefinitely upward even where the channels still have signal.

5 What the computation shows

Every altitude the instrument measures is diffusively separated or in the transition to it. The mean mass is already falling at 95 km, so the mixed regime lies below the range, and the assumption the retrieval makes is the right one over the range it claims.

The classical regime test cannot be run near the turbopause. Nitrogen and molecular oxygen have masses too close to the mean mass for a scale-height measurement to distinguish mixing from diffusion. Atomic oxygen, whose mass is far enough away, has a profile set by chemistry rather than by either transport process. The mean molecular mass is the better diagnostic.

A two-parameter temperature profile is accurate where it is used and wrong where it is not. One to two per cent above 200 km, 11 to 13 per cent at 130 km, and the boundary between those two statements is what sets the bottom of the retrieval's range.

6 What we decided

The retrieval assumes diffusive equilibrium on a Bates temperature profile, and section 4c is the justification rather than a convention inherited from prior work. The same section is the evidence for the altitude range over which the derived profile is claimed.

The nitrogen coverage limit reported in Unit 6 is restated as a property of the atmosphere rather than of the channel set. That changes what a reviewer should conclude from it: adding channels does not raise the limit, and a proposal claiming otherwise would be claiming something the physics does not allow.

Helium is omitted from the optical model, and this unit shows what that costs and where. Helium does not absorb in our channels, so it does not affect the measurement itself. It does affect the mass density derived from the species columns, above about 400 km, which is where helium starts to be a substantial fraction of the total mass. That is a statement about the derived product, not about the primary one.

7 What we do not claim

Not that NRLMSIS is the atmosphere. Every number in this unit comes from an empirical model, and Unit 1 cites a measurement finding that same model wrong by 26 per cent at dawn. This unit describes the structure the retrieval assumes, and the structure is well established; the particular values are a model's.

Not that the turbopause is a surface. It is a gradual transition over tens of kilometres in which the eddy and molecular diffusion coefficients cross, and it moves with season, latitude and activity. Treating it as an altitude is a convenience this unit uses and does not defend.

Not that the exobase figure is precise. It depends on a kinetic collision cross section taken as $3 \times 10^{-15}\,\mathrm{cm}^2$, a round number for a quantity that varies with species and temperature. A factor of two in that cross section moves the exobase by tens of kilometres. It is quoted to mark a regime boundary, not as a measurement.

Not that a quiet mid-latitude equinox profile represents the atmosphere. Unit 3 is about how much it moves, and every conclusion here should be read as conditional on that unit.

8 Exercises

E2.1 Recompute the mean molecular mass for a severe storm and for a winter pole. Which changes it more, and at which altitudes? Explain the sign of each change in terms of what heating does to the scale heights of a light species and a heavy one.

E2.2 Section 4b could not distinguish the mixed regime from the diffusive one using nitrogen, because nitrogen's mass is close to the mean mass. Identify a species pair and an altitude at which the test *would* be decisive, and say what measurement would be needed to perform it. Then say whether our instrument could make that measurement.

E2.3 Fit the Bates form to a storm temperature profile and repeat the table of section 4c. Does the exospheric temperature or the shape parameter carry the change? State what your answer implies for a retrieval that fits both from occultation data alone.

References

[1] J. T. Emmert, "Thermospheric mass density: a review," *Adv. Space Res.* **56**(5), 773–824 (2015), doi:10.1016/j.asr.2015.05.038.

[2] J. T. Emmert, D. P. Drob, J. M. Picone, D. E. Siskind, M. Jones Jr. et al., "NRLMSIS 2.0: a whole-atmosphere empirical model of temperature and neutral species densities," *Earth Space Sci.* **8**(3), e2020EA001321 (2021), doi:10.1029/2020EA001321.

[3] R. K. Sewell et al., "Thermospheric composition and temperature from GOES-R/SUVI solar occultations," *Space Weather* **23**, e2024SW004234 (2025); arXiv:2508.10242.

[4] G. R. Smith and D. M. Hunten, "Study of planetary atmospheres by absorptive occultations," *Rev. Geophys.* **28**(2), 117–143 (1990), doi:10.1029/RG028i002p00117.

Every quantitative result in this unit is provisional in the four ways that apply to the whole project: filter bands are idealized top-hats rather than measured transmission curves; the truth atmosphere is NRLMSIS 2.1; helium is omitted above 400 km; and no horizontal structure along the ray is modelled. Analysis code is in `channel-plan/`.